

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	2

Case Study Topic: DuckDB Columnar Storage using Skip-Index for Fast Filtering

Word Done Summary:

In this case study, we studied how DuckDB and column-oriented databases use skip-index or zone-map structures for fast filtering. We analysed how each block stores minimum and maximum values to avoid scanning unnecessary blocks during query execution. We also computed the storage overhead of the zone-map index and compared the performance of clustered and non-clustered filtering. Finally, we examined why columnar databases prefer clustered ingestion for improving query efficiency.

Implementation Details:

1. Studied the working of skip-index and zone-map structures.
2. Calculated storage overhead for block-level metadata.
3. Analysed block reads for random and clustered filtering.
4. Compared non-clustered and clustered query performance.
5. Evaluated asymptotic behaviour of skip-index effectiveness.
6. Studied the importance of sort-on-write in columnar systems.

Result:

The skip-index structure significantly reduces unnecessary block reads when data is clustered. Zone maps provide efficient filtering with very small storage overhead, making them highly effective for analytical database systems.

Output:

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DuckDB Skip-Index Analysis
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Total Rows: 1000000000
Rows Per Block: 10000
Number of Blocks: 100000
Skip-Index Storage: 1600000 bytes
Skip-Index Storage in MB: 1.52587890625

Conclusion:
Zone maps provide efficient filtering with very low storage overhead.
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GitHub Link:

<https://github.com/ketanasindhusomisetty/DSA-Case-Studies/tree/main>

Student Signature	Faculty Signature
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